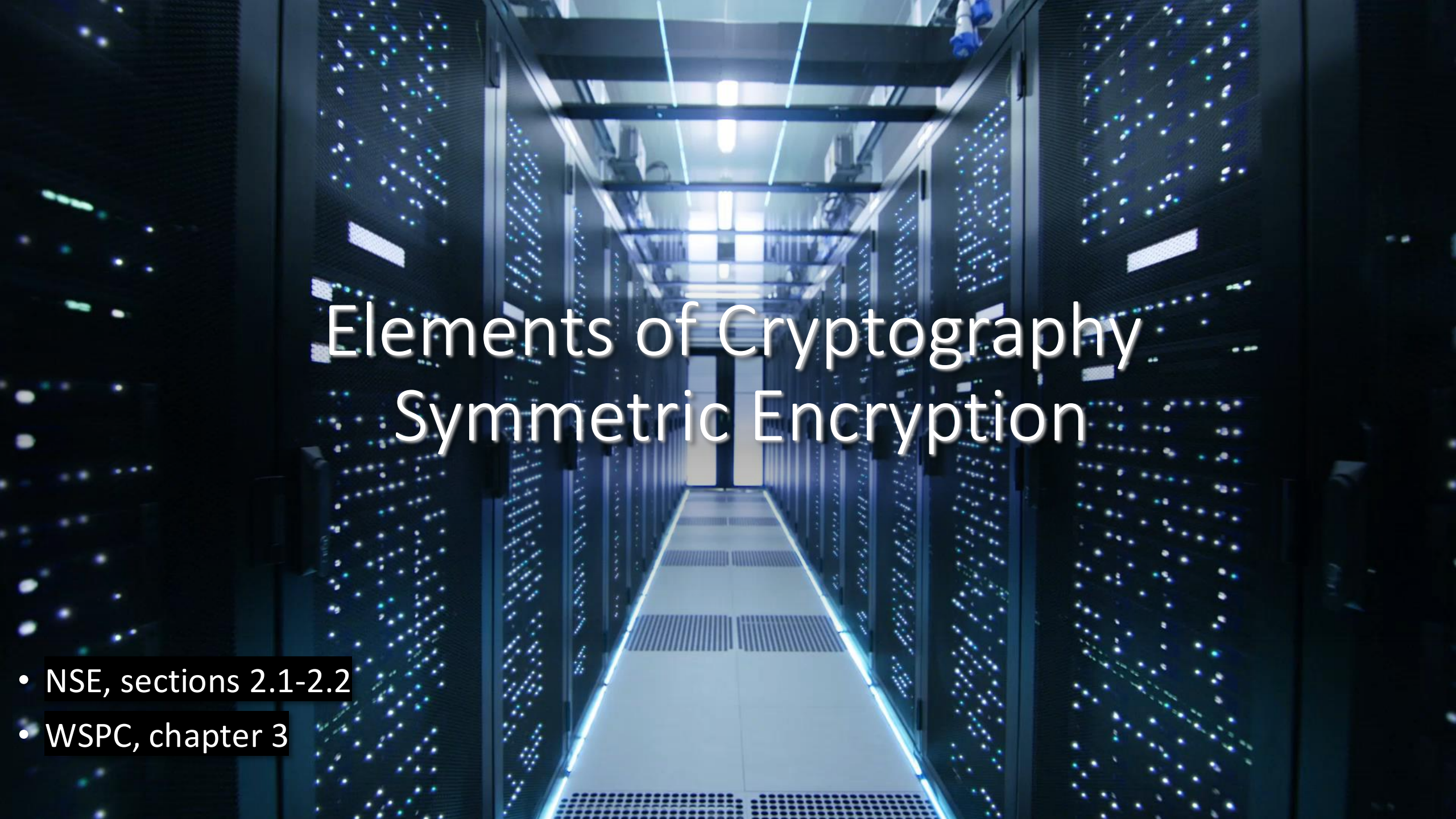




Elements of Cryptography

Symmetric Encryption

- NSE, sections 2.1-2.2
- WSPC, chapter 3

Cryptography



Cryptography is a collection of mathematical techniques for protecting information



Most important cryptographic technique is
encryption/decryption

Level	What to protect	Method
3	Existence of message	Steganography
2	Metadata of message	Privacy-enhancing technologies
1	Content of message	Encryption
0	Nothing	None

Table by I.A. Goldberg

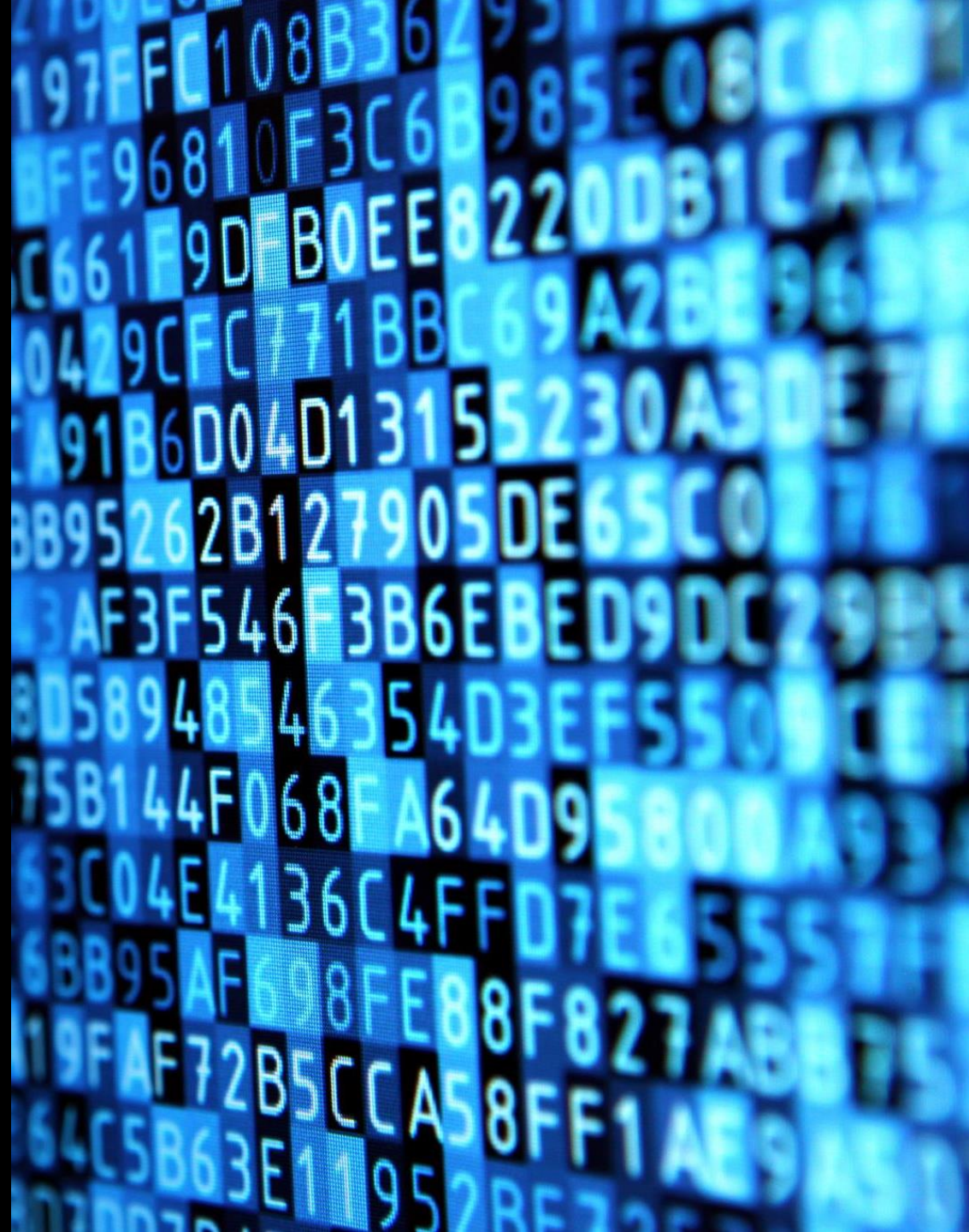
Cryptography for information protection

Encryption is used

- Directly at the level 1
- As an important ingredient at the levels 2 and 3

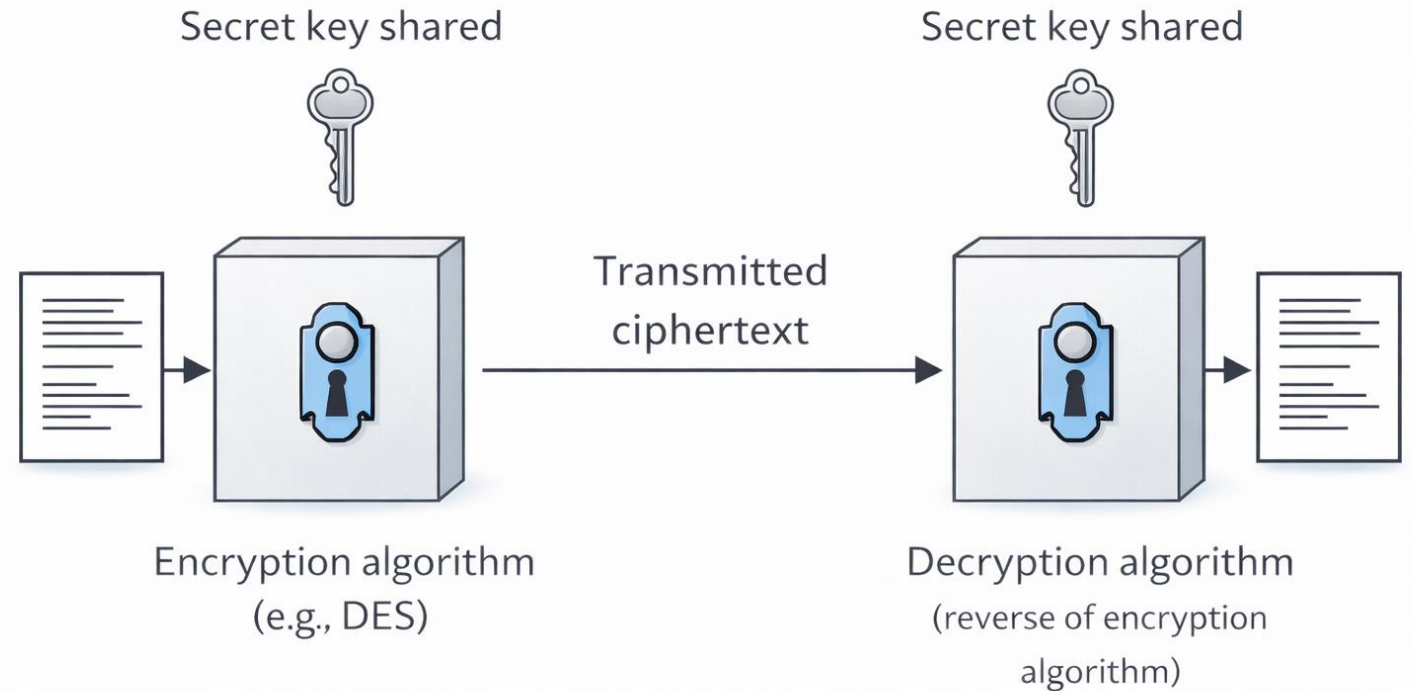
Two categories of encryption algorithms

- **Symmetric encryption** (or symmetric key encryption):
 - to encrypt and decrypt a message the *same key* (a piece of information; sequence of bits) is used
- **Asymmetric encryption** (or asymmetric key encryption):
 - One key is used for encryption (usually publicly known, *public key*);
 - Another key is used for decryption (usually *private*, or *secret key*)



Symmetric (conventional) encryption

Simplified symmetric key
encryption



Components of Symmetric Encryption

- Plaintext
- Encryption algorithm
- Secret key
- Ciphertext (encrypted text)
- Decryption algorithm

Security of symmetric encryption

Important principle

security of symmetric encryption depends on:

- the **secrecy of the key**,
- Not the secrecy of the algorithm

Why?

- It is **difficult to invent** new algorithms and *keep* them in a secret;
- Producing keys is much easier;



Requirements for symmetric encryption

- Strong encryption algorithm:
 - The adversary (opponent) should be unable to decrypt encrypted text, even if he/she knows several pairs
(plaintext, encrypted plaintext)
- Sender and receiver must have obtained copies of the secret key in a *secure way* and *must keep the key secure*

Two more classifications of cryptosystems

Type of operations used

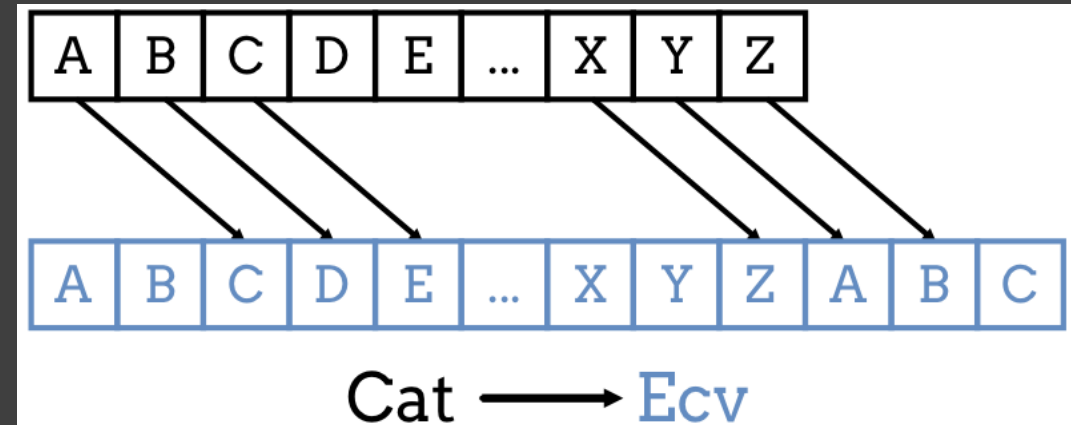
- Substitutions
- Transpositions

The way in which plaintext is processed

- **Block cipher:** input block of elements (e.g. characters) is transformed to the output block at once;
- **Stream cipher:** processes the input elements continuously, one element at a time.

Classics: substitutions

- Each element of the plaintext (bit, letter, group of bits) is mapped to another element



Classics: transposition

The idea behind a transposition cipher is to create a permutation (rearrangement) of the letters of the plaintext that will make the ciphertext appear to be well-encrypted.

Transposition ciphers are not highly secure because they do not change the letters in the plaintext or even cover up frequencies, but they can be built upon to make more secure methods of encryption.

Input: LIVERPOOL

Key Size = 3

L				R				L
	I		E		P		O	
		V				O		

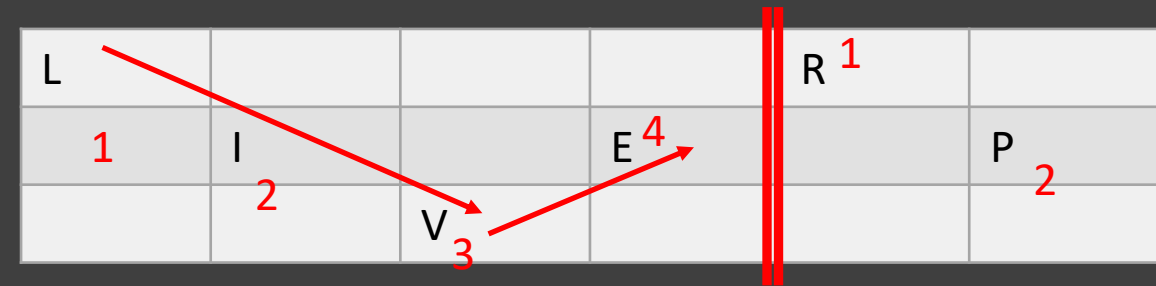
Output: LRLIEPOVO

Transposition – Rail Fence Cypher

- To solve the cipher, you must know the height and cycle of the puzzle. The height is simply the number of fence rails used to create it.
- In this example, we said that 3 fence rails were used, so the height is 3.
- The height will always be higher than 1 and no more than the number of letters in the ciphertext (in this case 9) or else the phrase will not be properly encoded; the height can thus be discovered by process of elimination if not known.

Rail Fence Cypher

- To determine the puzzle width, which will tell us how many total units will be in each row, you must determine the "cycle" of letters.
- A "cycle" of letters runs from the top row, down through each subsequent row, and then up again, but stopping before reaching the top row again. (The next letter on the top row will actually begin the next cycle.)
- Our 3-rail puzzle has a "cycle" of 4 letters



- So, divide the total number of characters (letters + spaces) by the cycle length to determine how many complete zig-zag cycles occur in the message, rounding down: $9 / 4 = 2.25$ (2)
- KEY =3 Cycle =4

LRLIEPOVO

L	1	2	3	(4)R	1	2	3	(4)L
---	---	---	---	------	---	---	---	------

IEPOVO

L				R				L
	I		E		P		O	

VO

L				R				L
	I		E		P		O	
		V				O		

UETNVRIYIS

Key=3

Cycle must be 4

UETNVRIYIS

U				E				T	
---	--	--	--	---	--	--	--	---	--

NVRIYIS

U				E				T	
	N		V		R		I		Y

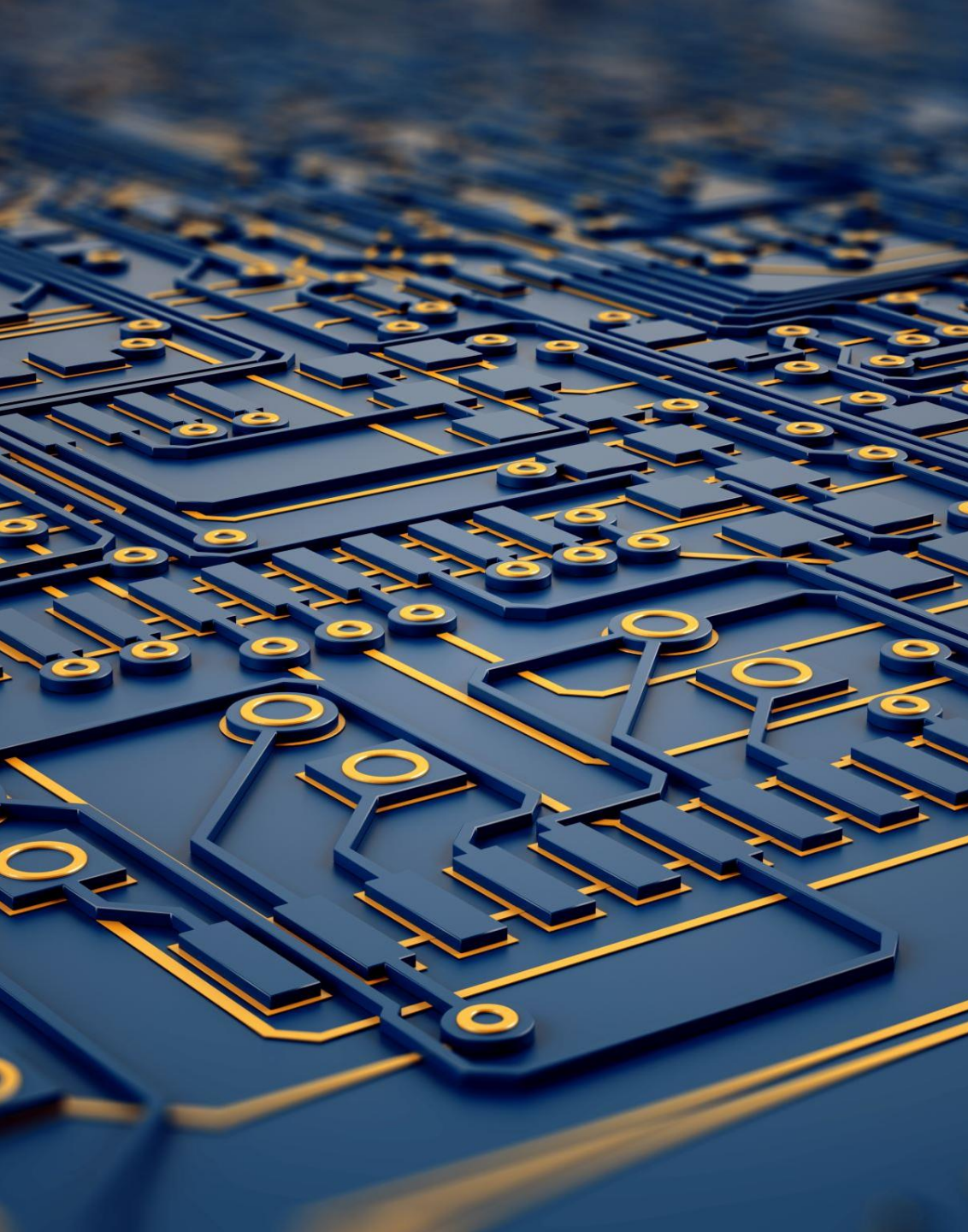
IS

U				E				T	
	N		V		R		I		Y
		I				S			

Two remarks

- Most modern algorithms include multiple stages of *interleaving* substitutions and transpositions;
- The encryption uses a large complex *key* (unlike simple examples on the previous slides)





Cryptanalysis and computationally secure schemes

- **Cryptanalysis:** The process of attempting to discover the plaintext or key;
- Depends very much on the information available;
- An encryption scheme is *computationally secure* if
 - The cost of breaking the scheme exceeds the value of the encrypted information;
 - The time required to break the scheme is more than lifetime of the information;

Table 2.1 Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext only	•Encryption algorithm •Ciphertext to be decoded
Known plaintext	•Encryption algorithm •Ciphertext to be decoded •One or more plaintext-ciphertext pairs formed with the secret key
Chosen plaintext	•Encryption algorithm •Ciphertext to be decoded •Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen ciphertext	•Encryption algorithm •Ciphertext to be decoded •Purported ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen text	•Encryption algorithm •Ciphertext to be decoded •Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key •Purported ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Types of Attacks (Cryptanalysis)

Dependent on the information known an attacker has several options

Brute-Force Approach in Cryptanalysis



If nothing else helps and there is no weakness in the encryption algorithms, brute-force approach may be applied;



Try every possible key until correct translation of the encrypted text into plaintext is obtained;



Possible issue: how does cryptanalyst recognize correct plaintext? Imagine it has been compressed before encryption;



Main issue: time !!

Key Size (bits)	Number of Alternative Keys	Time required at 1 encryption/ μs	Time required at 10^6 encryptions/ μs
32	$2^{32} = 4.3 \times 10^9$	$2^{31} \mu s = 35.8$ minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	$2^{55} \mu s = 1142$ years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	$2^{127} \mu s = 5.4 \times 10^{24}$ years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	$2^{167} \mu s = 5.9 \times 10^{36}$ years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	$2 \times 10^{26} \mu s = 6.4 \times 10^{12}$ years	6.4×10^6 years

A **microsecond** (μs) is an SI unit of time equal to one millionth (0.000001 or 10^{-6} or $1/_{1,000,000}$) of a second

Time required for brute-force search



Analysis vs Cryptanalysis

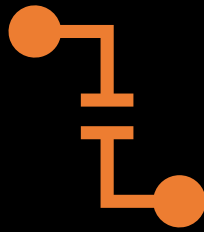
One of the main goals when creating new encryption algorithms is to make it as difficult as possible to break

Often simple algorithms provided the greatest security

But making an algorithm easy to analyse can be beneficial because:

Understanding can provide a high level of trust – ensuring no back doors or critical process flaws.

Block vs Stream



Block cipher

Input is blocks of elements – transformed to the output block all at once



Stream cipher

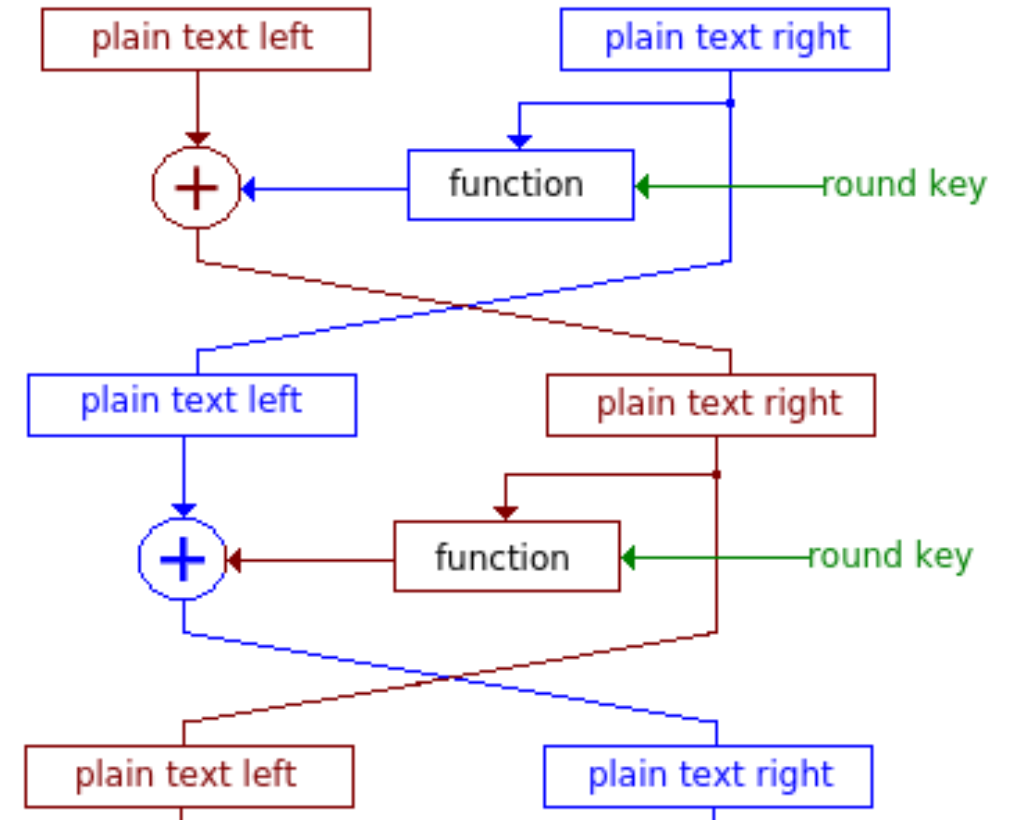
Process the elements continuously, one element at a time

Feistel cipher

- Most symmetric block algorithms have a structure similar to that proposed by H. Feistel in 1973
- The input is divided into the blocks of even numbers of elements
- Then multiple stage of substitutions and transpositions are applied
- Multiple keys (all derived from a master key) are used at the various rounds

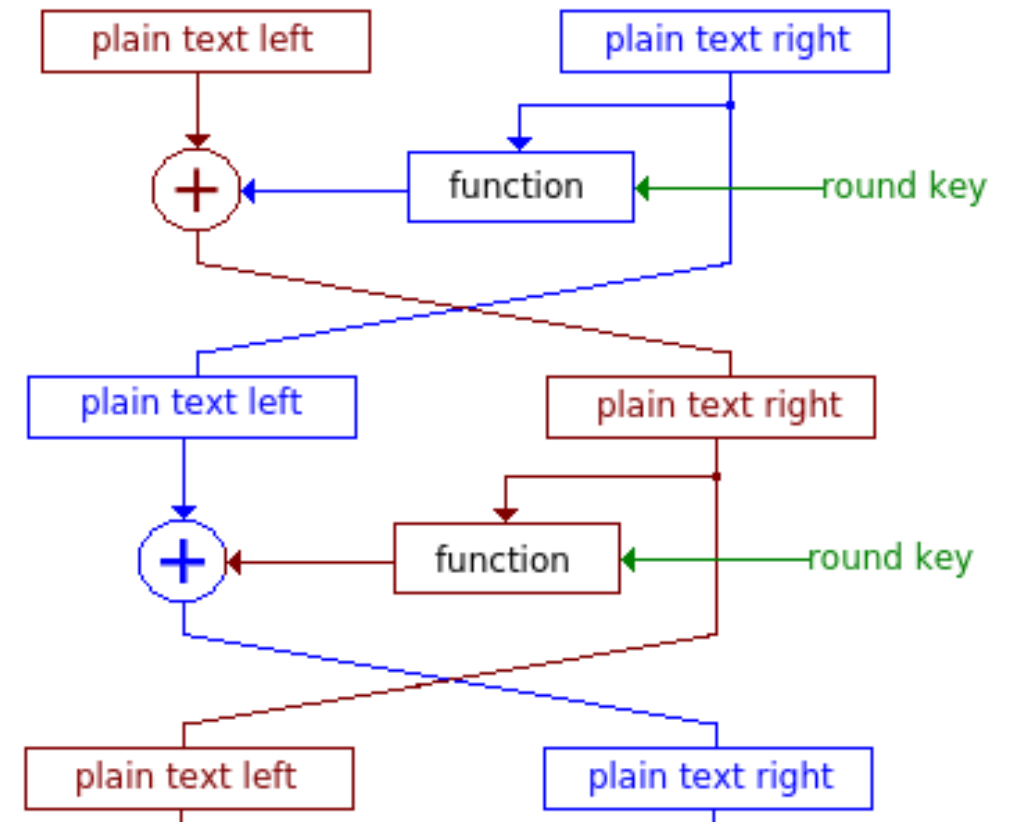
Feistel network

- Used all over cryptography
- A frame-work for building cryptography
- Does not have to be half and half exactly
- (balanced or imbalanced cipher)
- We take the RHS run it through a function and XOR with left and swap them over
- We then repeat this



Decryption

- Place the object to be decrypted back into the cipher and run the exact same process
- (But with reversed keys -even one way hash / functions).



Choices



Block size – Larger block size = more secure – but slower



Key size – Larger the size = more secure – but slower!



Number of rounds – more secure – but slower!



Sub key generation algorithm – a more complex algorithm generally means more security – may or may not influence speed



Round function – As above

Symmetric Encryption Algorithms



DES (Data Encryption Standard)



3DES (Triple DES)



AES (Advanced Encryption
Standard)

DES

- Adopted in 1977 by the National Bureau of Standards (now NIST)
- The Algorithm is DEA (Data encryption algorithm)
- Blocks – 64 bits
- Key – 56 bits
- Rounds – 16
- Sub keys – 16 (generated from master key)

Weakness of DEA

- Keys are only 56 bits
- Whilst 2^{56} is a large number computers can easily brute force this now (first in mid 90s)
- Solution: Increase the key size!

Strengths



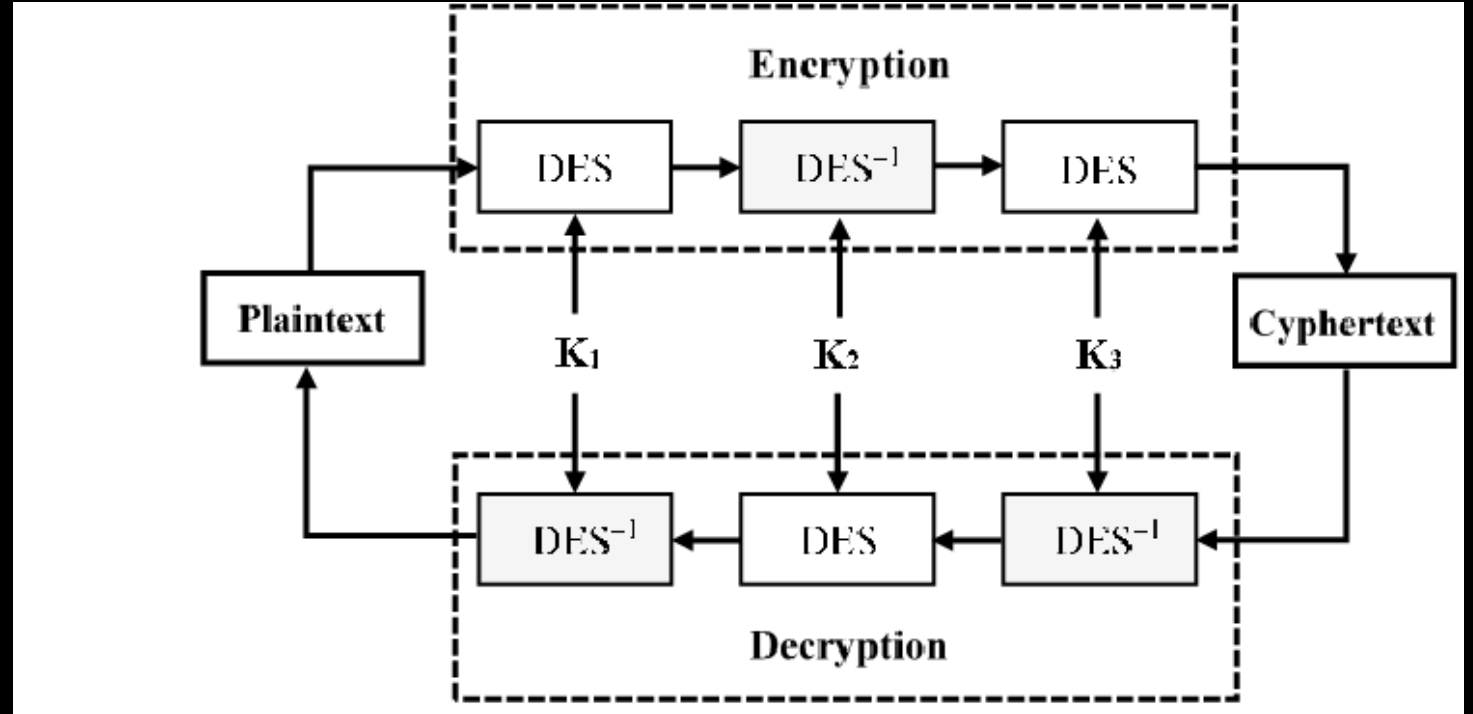
We do not know of a better way to attack this than brute force.



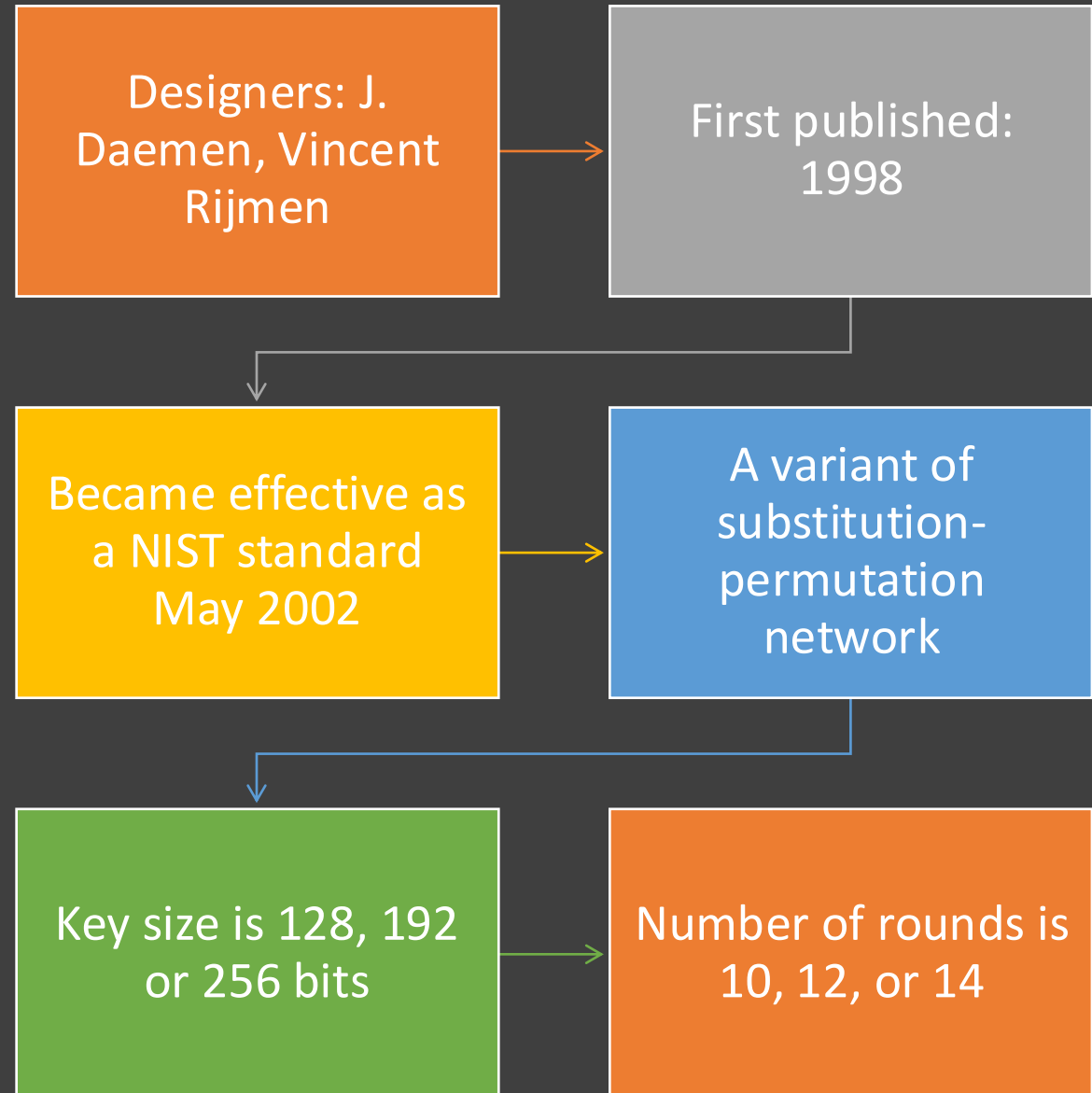
And this is only possible (in a reasonable time frame) if we use a weak key.

3DES

- All the power of DES but done 3 times
- Separate key for each execution – this also means more sub keys



Advanced Encryption Standard



Advanced Encryption Standard



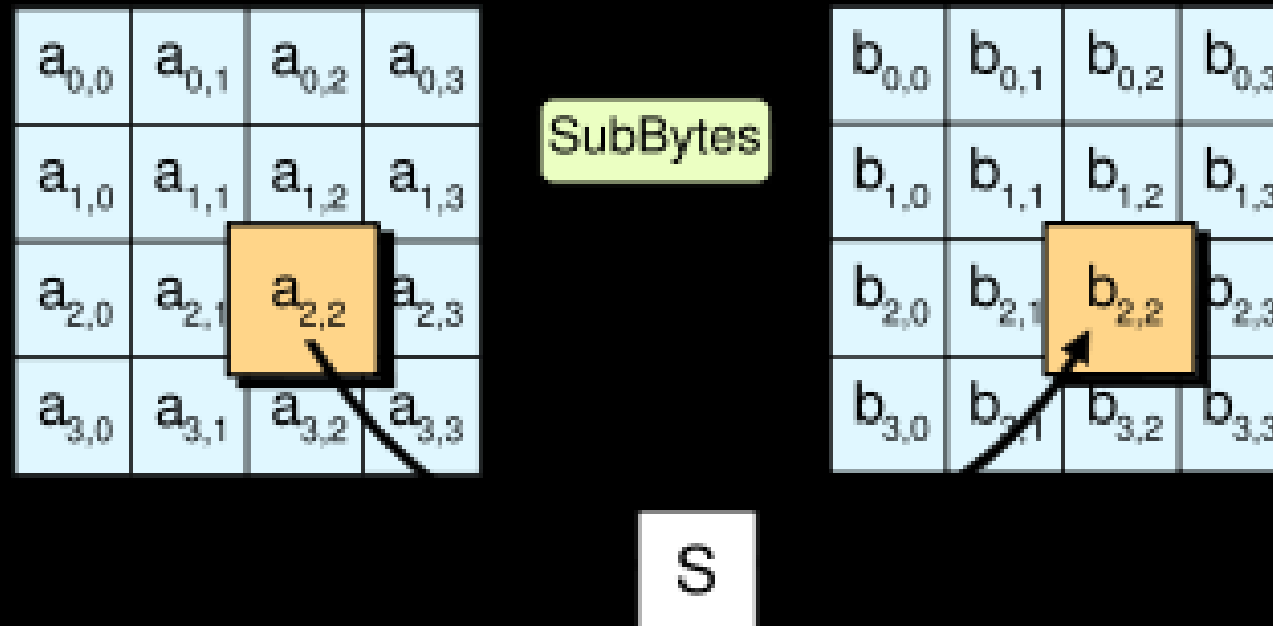
Design uses theory of finite fields, a branch of algebra;



Every block of 128 bits is presented as 4 by 4 array of bytes



Key Expansion: Key → Round keys



- Every round includes the following steps:
 - **Substitution**: each byte is replaced with another based on lookup table

Steps in AES processing, I

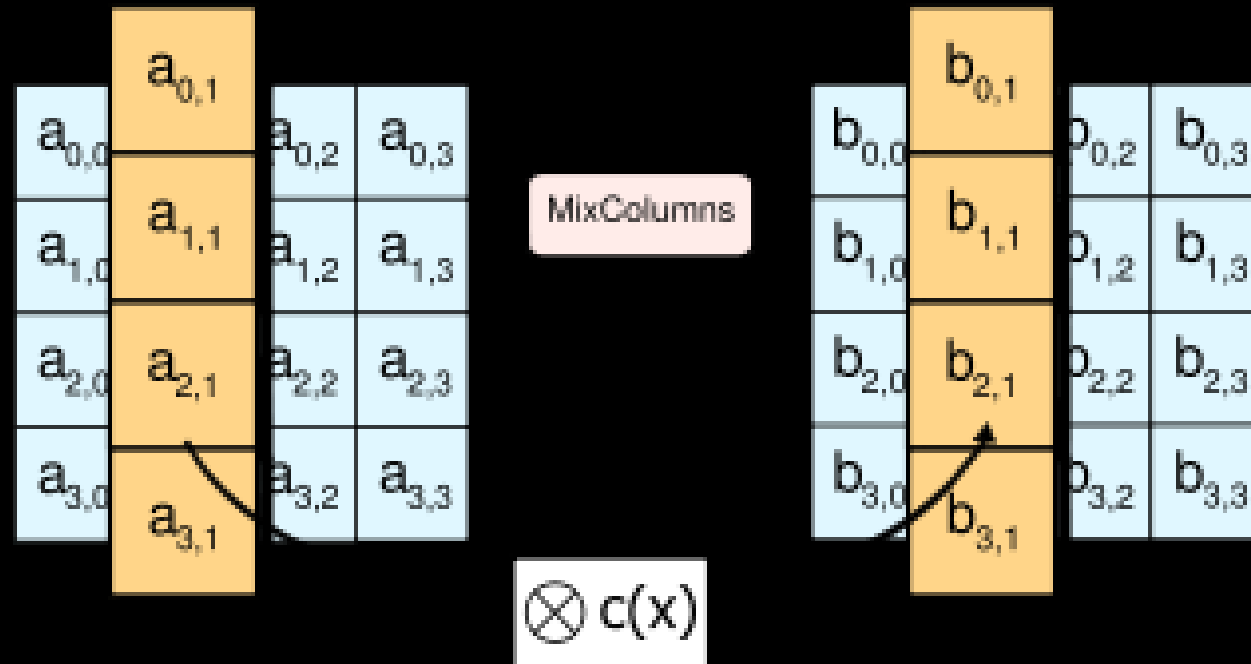
$a_{0,0}$	$a_{0,1}$	$a_{0,2}$	$a_{0,3}$
$a_{1,0}$	$a_{1,1}$	$a_{1,2}$	$a_{1,3}$
$a_{2,0}$	$a_{2,1}$	$a_{2,2}$	$a_{2,3}$
$a_{3,0}$	$a_{3,1}$	$a_{3,2}$	$a_{3,3}$

ShiftRows

$a_{0,0}$	$a_{0,1}$	$a_{0,2}$	$a_{0,3}$
$a_{1,1}$	$a_{1,2}$	$a_{1,3}$	$a_{1,0}$
$a_{2,2}$	$a_{2,3}$	$a_{2,0}$	$a_{2,1}$
$a_{3,3}$	$a_{3,0}$	$a_{3,1}$	$a_{3,2}$

- **ShiftRows:** each row is shifted cyclically certain amount of steps

Steps in AES processing, II

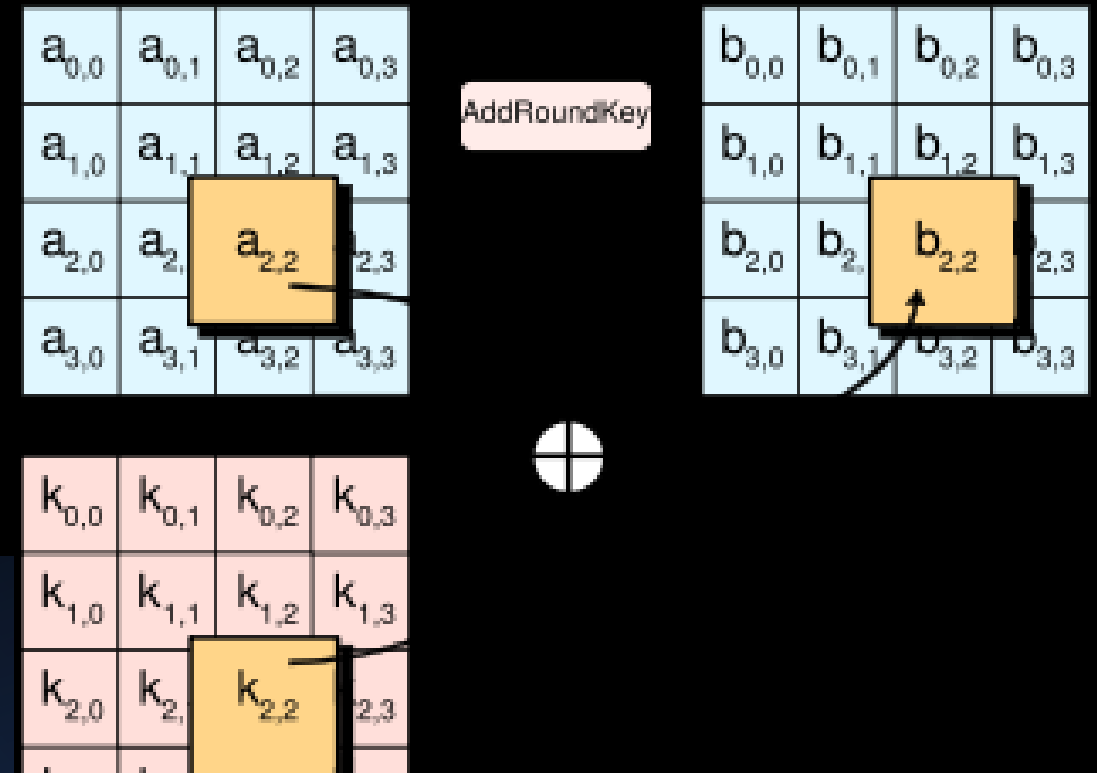


MixColumns: mixing operation on the columns (defined in terms of computations in a finite field).

Steps in AES processing, III

Steps in AES processing, IV

AddRoundKey: each byte is combined with the round key



Security of AES

Considered secure for use for classified information, secret and top-secret level

However, there are some concerns related to the **algebraic foundations** of algorithm – underlying algebraic structure might be used in the attacks in some clever way

The above is for Black Box setting; rather efficient **Side Channels attacks** have been discovered recently